Roll No. ....

TCS–101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

FUNDAMENTAL OF COMPUTER AND INTRODUCTION TO C PROGRAMMING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is flowchart ? Describe different types of flowcharts. Also explain the advantages of flowchart. What are the various symbols used in flowchart ? (CO1)

OR

(b) Explain each component of a computer with a neat block diagram. Also differentiate primary and secondary memory. (CO1)

2. Find the output with explanation : (CO2, CO4)

(i)
```
#include<stdio.h>
void main()
```

P. T. O.

```
{
   int x, y, z;
   x=1;y=2;z=3;
   if(x=4)
     z=x+y;
   printf("%d %d %d", x, y, z);
}
```

(ii)
```
#include<stdio.h>
void main()
{
   int a = 10, b;
   b = (a==10) ? 5:10;
   printf("%d", b);
}
```

(iii)
```
#include<stdio.h>
void main()
{
   int n = 50;
   while(n>0)
   {
     printf("%d", n%2);
     n/=2;
   }
}
```

(iv)
```c
#include<stdio.h>
void main()
{
    int m = 9, n = 0, p = 0;
    p+= n+= m;
    p+= n+= m+= -p;
    printf("%d %d %d", m, n, p);
}
```

(v)
```c
#include<stdio.h>
void main()
{
    int a = 15, b = 13, c = 16;
    int x, y;
    x = a - 3 % 2 + c * 2/4 % 2 + b/4;
    printf("%d %d", x, y);
}
```

3. (a) Explain unary, binary and ternary operators available in C. Also describe the role of precedence and associativity with an example. (CO2)

OR

(b) Explain in detail the data types, identifiers, keywords and constants of C. (CO2)

4. (a) Draw a flowchart to input co-ordinates of a point (x, y) and display a message in which quadrant the point will lie. (CO3)

OR

(b) Draw a flowchart to find minimum of N different integer numbers. (CO3)

5. (a) Write a C program to calculate and print the electricity bill of a given customer. Input unit and calculate the bill according to following criteria : (CO4)

Unit	Charge per Unit
Upto 199	@ 1.20
200 and above and less than 400	@ 1.50
400 and above and less than 600	@ 1.80
600 and above	@ 2.00

OR

(b) Write a C program to input a three-digit number. Swap its first and last digit. Print the final number. (CO4)

Roll No. ....

TMA–101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ENGINEERING MATHEMATICS—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Find rank of the matrix $\begin{bmatrix} 4 & -1 & 5 \\ 2 & 4 & -6 \\ 1 & 3 & 5 \end{bmatrix}$ using normal form. (CO1)

OR

(b) Determine whether the following set of vectors is linearly dependent or independent. (CO1)

$$\{(1, 1, 1), (1, 2, 3), (2, 3, 4)\}$$

2. (a) For what value of λ and μ, the following system of equations : (CO1)

$$2x+3y+5z=9$$

$$7x+3y-2z=8$$

$$2x+3y+\lambda z=\mu$$

P. T. O.

will have :

(i) Unique solution

(ii) Infinite solutions

(iii) No solution ?

OR

(b) Find the non-singular matrix P and Q so that PAQ is a Normal form where : (CO1)

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & 1 \end{bmatrix}$$

3. (a) Find Eigen values and Eigen vectors of the matrix : (CO1)

$$\begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 4 & 0 & 6 \end{bmatrix}$$

OR

(b) If : (CO1)

$$u = \log\left(\frac{x^4 - y^4}{x - y}\right)$$

prove that :

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -3.$$

4. (a) If : (CO2)

$$z = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$$

prove that :

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 2z.$$

OR

(b) If : (CO2)

$$x^x y^y z^z = c$$

Show that at $x = y = z, \frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$.

5. (a) Expland $\cos x \cos y$ near the point (0, 0) by Taylor's series upto third degree term. (CO2)

OR

(b) Expand $e^x \cos y$ near the point $\left(1, \frac{\pi}{4}\right)$ upto second degree terms.

(CO2)

Roll No. ....

TCH–101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

(All Branches)

ENGINEERING CHEMISTRY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Using molecular orbital theory, compare the bond energy and magnetic character of N_2, N_2^+ and N_2^- species. (CO1)

OR

(b) Explain Band Theory. Discuss extrinsic and intrinsic semiconductors. (CO1)

2. (a) (i) Differentiate between bonding and antibonding molecular orbitals.

(ii) Differentiate between intermolecular and intramolecular hydrogen bonding (CO1)

OR

(b) Explain different types of transitions involved in UV spectroscopy. Write applications of UV spectroscopy. (CO1)

P. T. O.

3. (a) What is zeolite ? Explain zeolite process for water softening with its benefits, and limitations. (CO2)

OR

(b) Calculate the amount of lime (85% pure) and soda (90 % pure) required for treatment of 20,000 litres of water containing : (CO2)

$Mg(HCO_3)_2 = 18.2,\ CaCl_2 = 13.8, Ca(HCO_3)_2 = 20.3,$

$MgCl_2 = 17.0, MgSO_4 = 15.0,\ CaSO_4 = 25.0$ and $NaCl = 25$

4. (a) Write notes on the following : (CO2)

(i) Reverse osmosis

(ii) Caustic embrittlement

OR

(b) (i) A completely exhausted zeolite softener requires 80 liters of NaCl solution having 100 g/l of NaCl. Calculate the hardness of water if 1500 liters of water was softened by the zeolite softener. (CO2)

(ii) Discuss boiler problems due to hard water.

5. (a) Explain lime soda process used for softening of hard water by giving suitable reactions. (CO2)

OR

(b) What are the advantages and disadvantages of use of hard water ?

A water sample contains 408 mg of $CaSO_4$ per liter. Express the hardness in $CaSO_3$ equivalents in ppm, °French, °Clark and mg/L.

(CO2)

Roll No. ....

TPH–101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

(All Branches)

ENGINEERING PHYSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the Newton's ring experiment. Also derive that in Newton's ring experiment, the spacing between the rings decreases with the order of the ring. (CO1)

OR

(b) In Newton's ring experiment the diameter of 4th and 12th dark rings are 0.400 cm and 0.700 cm respectively. Deduce the diameter of 20th dark ring and radius of curvature of plano-convex lenx (wavelength 5893 Å). (CO1)

2. (a) Distinct the Fresnel and Fraunhofer diffraction. Show that for a single slit Fraunhofer diffraction the relative intensities of the principal, the first, the second maximum....... are $1, 4/9\pi^2; 4/25\pi^2, .4/49\pi^2$....... etc..

P. T. O.

OR

(b) A single slit is illuminated by light composed of two wavelengths λ_1 and λ_2. One observes that the due to Fraunhofer diffraction the first minima obtained for λ_1 coincides with the second diffraction minima of λ_2. What is the relation between λ_1 and λ_2 ? Derive the Resolving Power of a grating. (CO1)

3. (a) Explain the construction and working of He-Ne laser. How is it superior to a ruby laser ? (CO2)

OR

(b) Find the intensity of a laser beam of 1 mW power and having a diameter of 1.4 mm. Assume the intensity to be uniform across the beam. (CO2)

4. (a) Explain Fresnel's biprism with a diagram and derive the expression to calculate the distance between two sources using the deviation method. (CO1)

OR

(b) Find the minimum number of lines in a diffraction grating required to just resolve the sodium doublet (5890 Å and 5896 Å) in the (i) First Order (ii) Second Order. (CO1)

5. (a) Explain the stimulated emission in the laser. Derive the expression for the relation between Einstein's Coefficients. (CO2)

OR

(b) The coherence length of sodium light is 2.945×10^{-2} m and its wavelength is 5890Å. Calculate the (i) frequency (ii) coherence time. (CO2)

Roll No. ....

TEE–101

B. TECH. (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

(All Branches)

BASIC ELECTRICAL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the following : (CO1/CO2)

(i) Branch

(ii) Loop

(iii) Mesh

(iv) Node

(v) Junction

(vi) Current

(vii) EMF

(viii) Potential

(ix) Circuit

(x) Network

P. T. O.

OR

(b) What is KCL ? Find the voltage at node 1 and 2 for the network shown in Fig. 1 below using nodal analysis. (CO1/CO2)

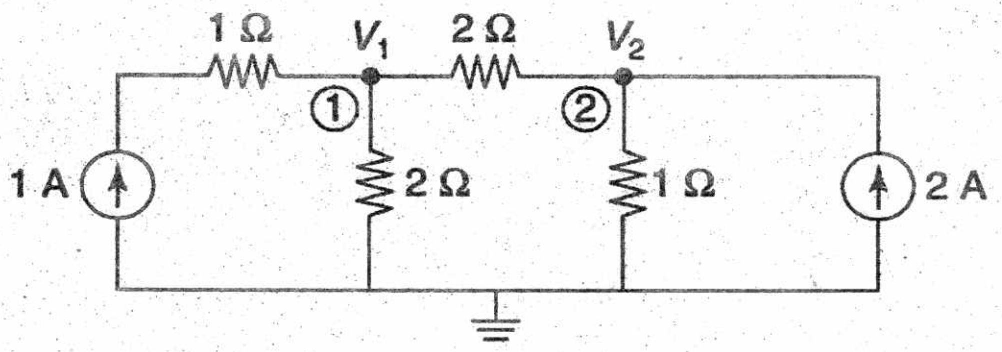

Fig. 1

2. (a) Classify and explain the different types of network elements and different types of network using suitable example. (CO1/CO2)

OR

(b) Find the current flowing through 2 ohm resistance using mesh analysis in the Fig. 2. (CO1/CO2)

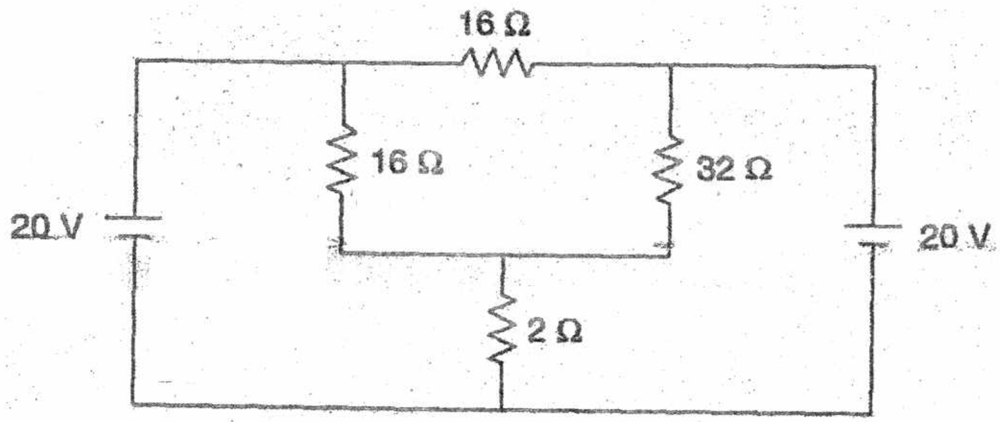

Fig. 2

3. (a) Find the current through 2 ohm resistor connected between terminal A and B in the Fig. 3, using Thevenin's theorem. Also state the Thevenin's theorem. (CO1/CO2)

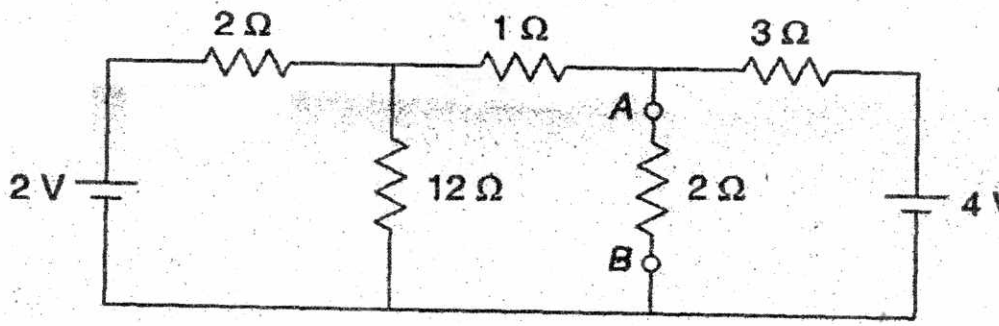

Fig. 3

OR

(b) Explain and classify the different independent energy source. Also draw the I-V characteristic of the energy sources. (CO1/CO2)

4. (a) State superposition theorem and the conditions when it can be applied. Find the current in 10 ohm resistor in circuit shown in Fig. 4, using superposition theorem. (CO1/CO2)

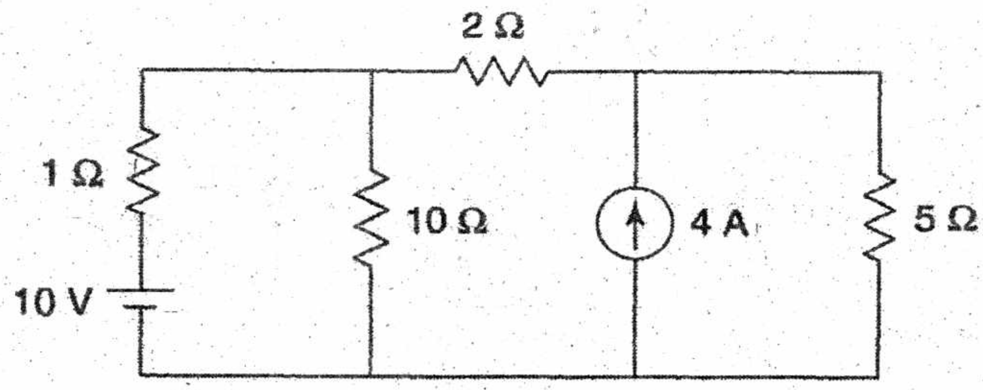

Fig. 4

OR

(b) Explain the following terms :

(i) Waveform

(ii) Cycle

(iii) Frequency

(iv) Time period

(v) Amplitude

(vi) Average value

(vii) Instantaneous value (CO3)

5. (a) Find the equivalent resistance between A and B in the network shown in Fig. 5. (CO1/CO2)

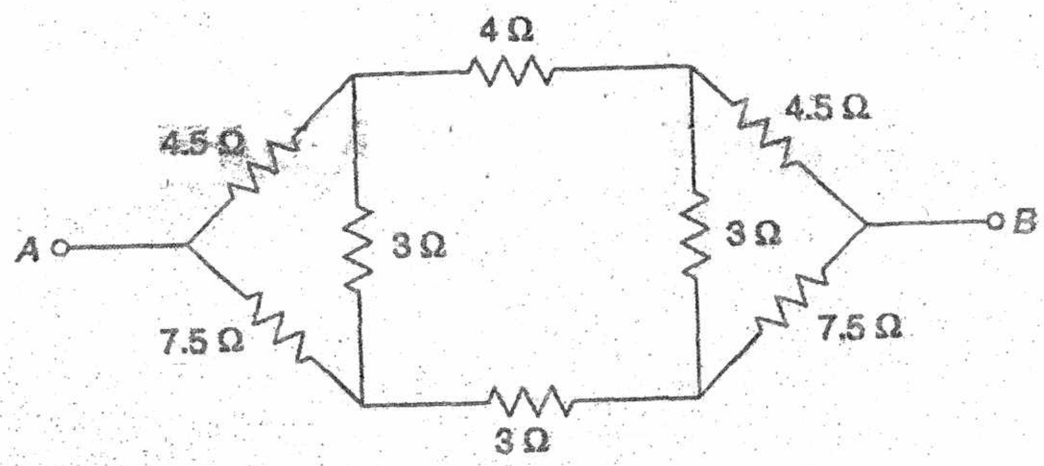

Fig. 5

Roll No. ....

TEC–101

B. TECH. (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

(All Branches)

BASIC ELECTRONICS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve : (CO1)

(i) $(B9A95.0C)_{16} = (?)_{10}$

(ii) $\sqrt{(41)_x} = (5)_x$, find the base "x".

(iii) $(63542)_7 = (?)_{10}$

(iv) $(7452376)_8 = (?)_{16}$

(v) $(9947.562)_{10} = (?)_7$

OR

(b) Simplify the following expressions with the help of Boolean algebra : (CO1)

(i) $F = AC + AD + AC'D' + AB$

(ii) $F = AB'C + A'B'C + A'B'C' + AB'C'$

(iii) $F = AB + A(B + C) + B(B + C)$

(iv) $F = \left[AB'(C + BD) + A'B'\right] C$

(v) $F = A + AB + AB'C$

P. T. O.

2. (a) Minimize the following by K map and realize the minimized function by NOR gate only : (CO1)

$$F(A,B,C,D,) = m(1,3,7,11,15) + d(0,2,5)$$

OR

(b) Simplify the following function with the help of K Map : (CO1)

$$F(A, B, C, D,) = B'D + A'D + BD$$

3. (a) What is energy band diagram ? Explain classification of materials on the basis of energy band diagram. (CO2)

OR

(b) If N_D and N_A are donor and acceptor impurities and n_i is the intrinsic concentration, establish the relation for minority and majority charge densities in extrinsic semiconductors. (CO2)

4. (a) A sample of Ge is made *p*-type by adding acceptor atoms at a rate of one atom per $4 \times 10^8 / m^3$. (CO2)

If density of Ge atoms is $4.4 \times 10^{28} / m^3$, determine electron and hole concentration in the doped semiconductor if $n_i = 2.5 \times 10^{19} / m^3$.

OR

(b) Calculate the conductivity and resistivity of pure silicon at room temperature when the concentration of carrier is $1.6 \times 10^{10} / cm^3$. (CO2)

Given that the mobility of electron is $1500\ cm^2 / V\text{-sec}$ and the mobility of holes is $1500\ cm^2 / V\text{-sec}$.

5. (a) Calculate the density of impurity atom that must be added to an intrinsic Si crystal to convert it to a : (CO1/CO2)

(i) $10^4 \Omega m$ N-type silicon

(ii) $10^4 \Omega m$ P-type silicon

$\mu_n = 0.138\ m^2 / V\text{-sec}$, and $\mu_p = 0.046\ m^2 / V\text{-sec}$

OR

(b) Prove the following theorems : (CO1/CO2)

(i) $A + (BC) = (A + B)(A + C)$

(ii) $A + A'B = A + B$

Roll No. ....

TCE–101

B. TECH. (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

(All Branches)

BASIC CIVIL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the term "Civil Engineering" and outline various sub-disciplines within civil engineering. (CO1)

OR

(b) Enlist the *ten* reasons to demonstrate the significant role that infrastructure plays in the nation's economic advancement. (CO1)

2. (a) List out any *five* civil engineering marvels in the world highlighting their key civil engineering features. (CO1)

OR

(b) Describe the following structures :

(i) Tower

(ii) Retaining walls

(iii) Dams

(iv) Railway (CO1)

P. T. O.

3. (a) Write and explain the concept of "whole to part" in respect of surveying. (CO2)

OR

(b) Write a short note on various instruments used in linear measurement in surveying. (CO2)

4. (a) Write a short note on plane surveying and geodetic surveying. (CO2)

OR

(b) Enlist the various types of surveying techniques on the basis of place of survey. Explain any *two* types. (CO2)

5. (a) Convert WCB into QCB for the following :

(i) 186°

(ii) 129° (CO2)

OR

(b) Explain the concept and working of GPS with the help of a neat diagram. (CO2)